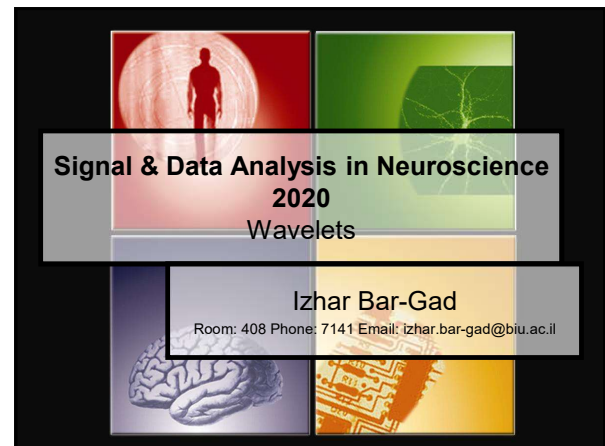
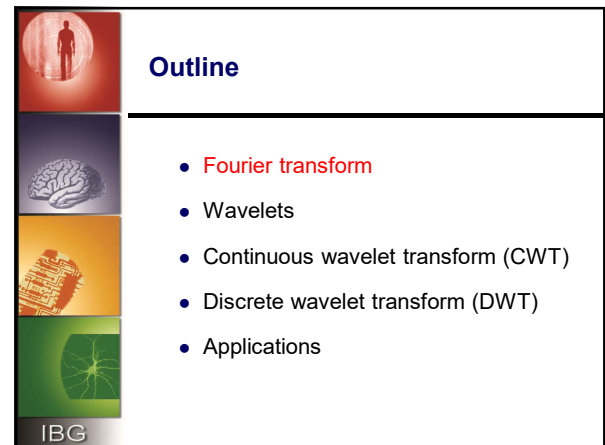



**Signal & Data Analysis in Neuroscience
2020
Wavelets**

Izhar Bar-Gad
Room: 408 Phone: 7141 Email: izhar.bar-gad@biu.ac.il

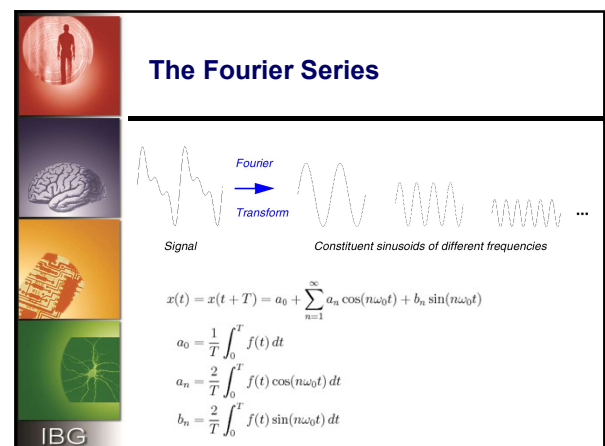
1



Outline

- **Fourier transform**
- Wavelets
- Continuous wavelet transform (CWT)
- Discrete wavelet transform (DWT)
- Applications

2



The Fourier Series

Signal $\xrightarrow{\text{Fourier Transform}}$ Constituent sinusoids of different frequencies

$$x(t) = x(t + T) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)$$

$$a_0 = \frac{1}{T} \int_0^T f(t) dt$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos(n\omega_0 t) dt$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin(n\omega_0 t) dt$$

3



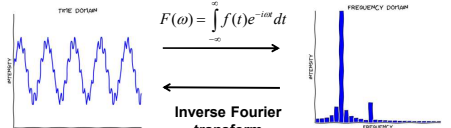
The Fourier transform

Transform the view of the signal from the time domain to the frequency domain.

Fourier transform

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt$$

Inverse Fourier transform

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega)e^{i\omega t} d\omega$$


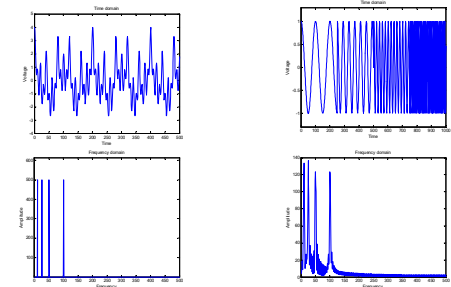
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4



No temporal precision

$x(t) = \cos(2\pi \cdot 10t) + \cos(2\pi \cdot 25t) + \cos(2\pi \cdot 50t) + \cos(2\pi \cdot 100t)$
 $x(t) = [\cos(2\pi \cdot 10t), \cos(2\pi \cdot 25t), \cos(2\pi \cdot 50t), \cos(2\pi \cdot 100t)]$



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5



Disadvantages of Fourier transform

- Some signals have spectral characteristics that vary with time.
- By using FT we lose the time information: WHEN did a particular event take place?
- The Fourier transform can not locate drift, trends, abrupt changes, beginning and ends of events, etc.

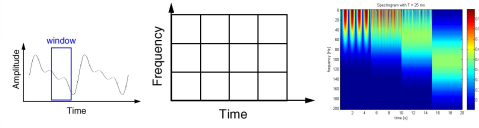
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6



Short Time Fourier Transform (STFT)

$$\text{STFT}\{x(\cdot)\} \equiv X(\tau, \omega) = \int_{-\infty}^{\infty} f(t)w(t-\tau)e^{-i\omega t} dt$$

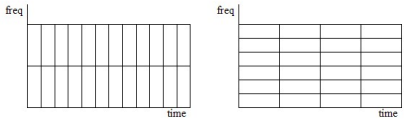
$$\text{spectrogram}\{x(\cdot)\} \equiv |X(\tau, \omega)|^2$$


7



The Heisenberg uncertainty principle

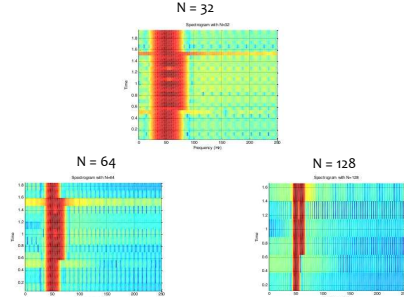
Trade off: Higher frequency resolution means lower time resolution, and vice versa.



8



Time-frequency trade off



9






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Disadvantages of the STFT

- A compromise between time-based and frequency-based views of a signal.
- Both time and frequency are represented in limited precision (window size based).
- Once you choose a particular size for the time window - it will be the same for all frequencies.

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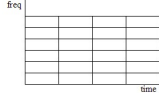





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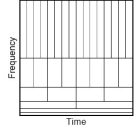
Another solution...

The STFT



Problem:
Constant time/frequency resolution

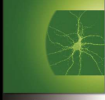
Wavelet transform



Solution:
Varying time/frequency resolution

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Outline

- Fourier transform
- **Wavelets**
- Continuous wavelet transform (CWT)
- Discrete wavelet transform (DWT)
- Applications

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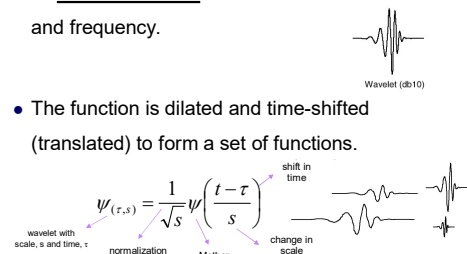
Wavelets

- A “**mother wavelet**” is a function located in time and frequency.
- The function is dilated and time-shifted (translated) to form a set of functions.

Wavelet (db10)

$$\psi_{(\tau,s)} = \frac{1}{\sqrt{s}} \psi\left(\frac{t-\tau}{s}\right)$$

Annotations:
- $\psi_{(\tau,s)}$: wavelet with scale, s and time, τ
- $\frac{1}{\sqrt{s}}$: normalization
- ψ : Mother wavelet
- $\frac{t-\tau}{s}$: shift in time and change in scale



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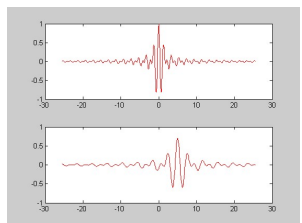

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Example

Shannon wavelet: $Y(t) = 2 \text{sinc}(2t) - \text{sinc}(t)$

Mother wavelet

$t=5, s=2$



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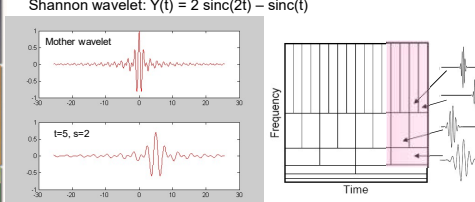

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Example

Shannon wavelet: $Y(t) = 2 \text{sinc}(2t) - \text{sinc}(t)$

Mother wavelet

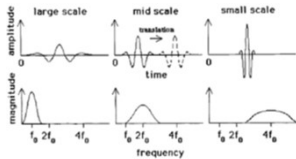
$t=5, s=2$



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Time-frequency Multi-resolution analysis



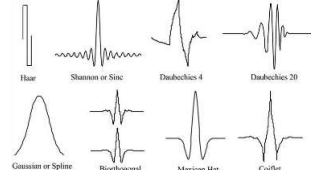
(Samar et al., 1999)

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Wavelets Characteristics

- A wavelet is a waveform of effectively limited duration.
- Finite L_1 and L_2 norms $\int_{-\infty}^{\infty} |\psi(t)| dt < \infty$ and $\int_{-\infty}^{\infty} |\psi(t)|^2 dt < \infty$



$$\int_{-\infty}^{\infty} \psi(t) dt = 0 \quad \int_{-\infty}^{\infty} |\psi(t)|^2 dt = 1$$

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Outline

- Fourier transform
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Continuous wavelet transform (CWT)

STFT

$$X(\omega, \tau) = \int_{-\infty}^{\infty} f(t) w(t - \tau) e^{-i\omega t} dt$$

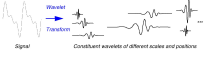
Frequency, time analyzing function



Wavelet transform

$$X(s, \tau) = \int_{-\infty}^{\infty} f(t) \frac{1}{\sqrt{s}} \psi\left(\frac{t - \tau}{s}\right) dt$$

scale, translation analyzing function



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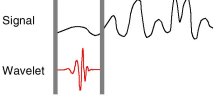
19






How does it work?

- Step 1:** Take a Wavelet and compare it to a section at the start of the original signal.



Signal

Wavelet

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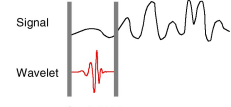
20






How does it work?

- Step 2:** Calculate the coefficient, C, that represents how closely correlated the wavelet is with this section of the signal. The higher C is, the more the similarity.



Signal

Wavelet

C = 0.0102

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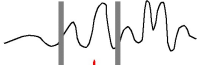



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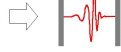
How does it work?

- **Step 3:** Shift the wavelet to the right and repeat steps 1-2 until you have covered the whole signal.

Signal



Wavelet



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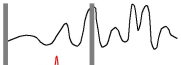



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How does it work?

- **Step 4:** Scale (stretch) the wavelet and repeat steps 1-3.

Signal



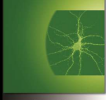
Wavelet



$C = 0.2247$

23

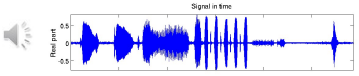


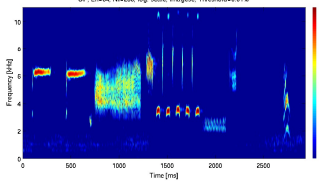
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Sparrow's song Time-frequency representation

Signal in time



SP: L1=64, N1=256, log scale, images: Threshold=0.01%

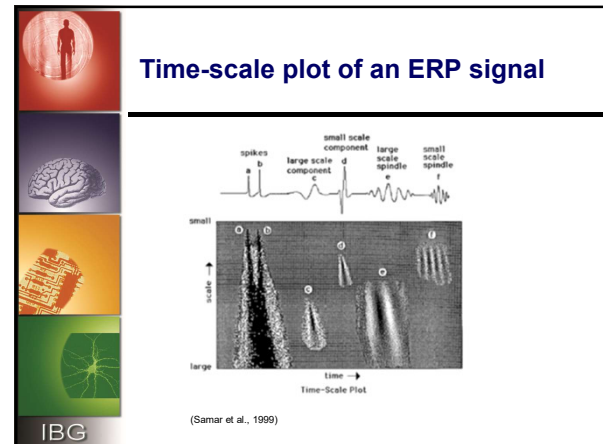


Frequency [Hz]

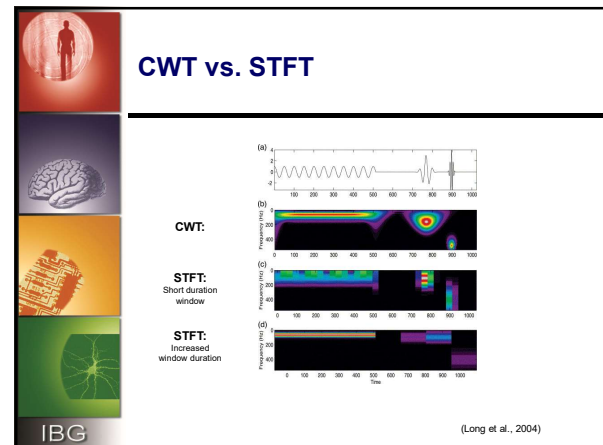
Time [ms]

http://tbb.nongnu.org/demos_ft.html

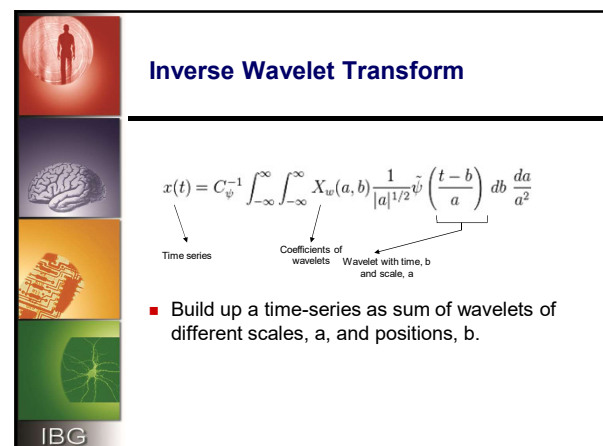
24



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Disadvantages of the CWT

- There are an infinite amount of coefficients/wavelets.
- Efficiency – The information displayed at closely spaced scales or at closely spaced time points is highly correlated and therefore unnecessarily redundant for many analytic purposes.
- Time consuming to compute directly.

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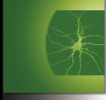

Outline

- Fourier transform
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- Applications

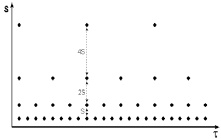
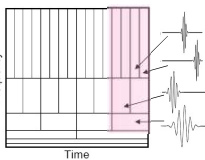
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DISCRETE WAVELET TRANSFORM (DWT)

$$\psi_{(j,k)} = \frac{1}{\sqrt{2^j}} \psi\left(\frac{t - k \cdot 2^j}{2^j}\right)$$



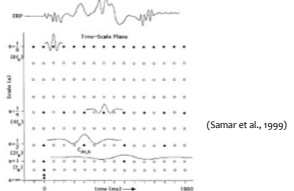
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DWT vs. CWT

- Only the discrete set is necessary to completely represent the ERP.
- The CWT can produce thousands of coefficients for a single waveform.
- The DWT produces only as many coefficients as there are samples within the original signal, without the loss of any information at all.
- The DWT is not time-invariant, it is very sensitive to the alignment of the signal in time.

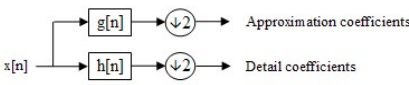


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DWT - series of filters

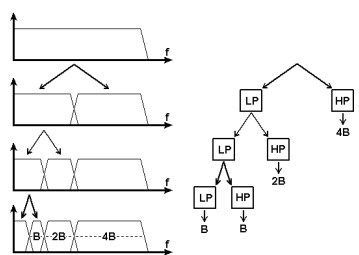
- There are two filters involved:
 - "Wavelet filter" - high pass filter.
 - "Scaling filter" - low pass filter.



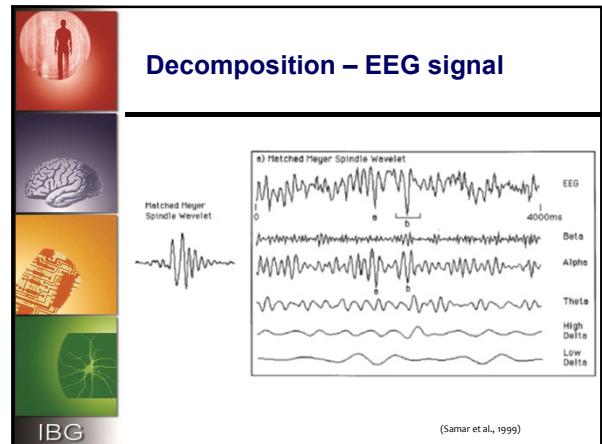
32



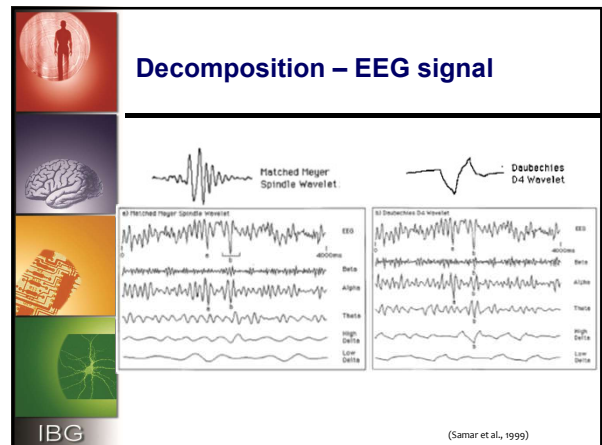
Wavelet decomposition tree Filter bank



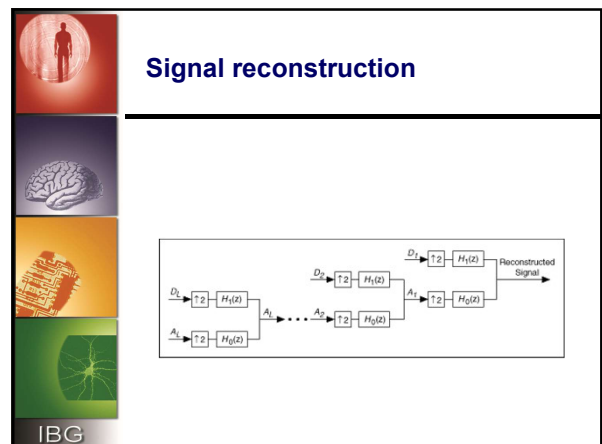
33



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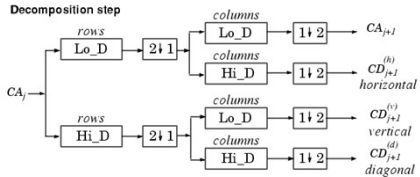


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Two-Dimensional DWT

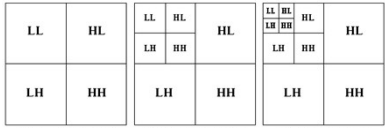
Decomposition step



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Two-Dimensional DWT



(a) Single Level Decomposition (b) Two Level Decomposition (c) Three Level Decomposition

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Outline

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Applications

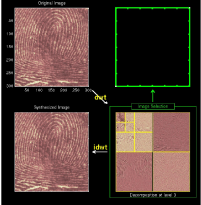
- EEG/ERP
- Spike detection
- 2D Images
- HVS detection

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Compression

- A two dimensional (image) compression, using 2D wavelets analysis.
- The FBI uses a wavelet technique to compress its fingerprints database.



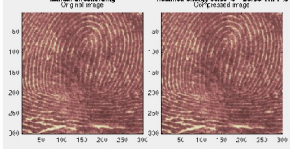
41



Finger print compression

Original Image

Compressed Image



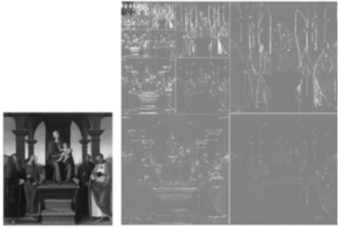
Threshold: 3.5
Zeros: 42%
Retained energy: 99.95%

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Art Authentication



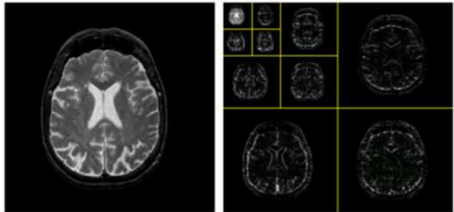

Lyu et al., PNAS rev, 2004

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MRI





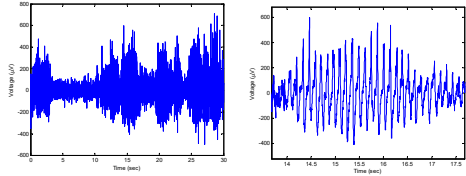
44





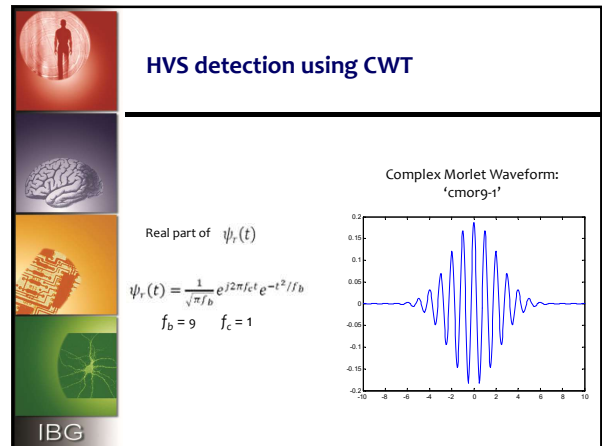

High voltage spindles (HVS) detection

HVSs appear in awake immobile rats and are characterized by atypical frequency of 5-13Hz with atypical spike and wave shape.

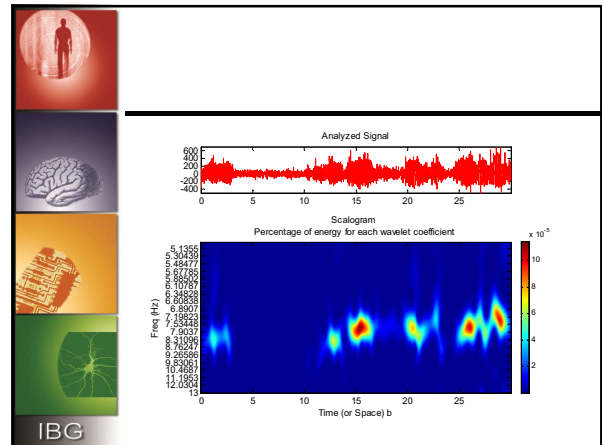




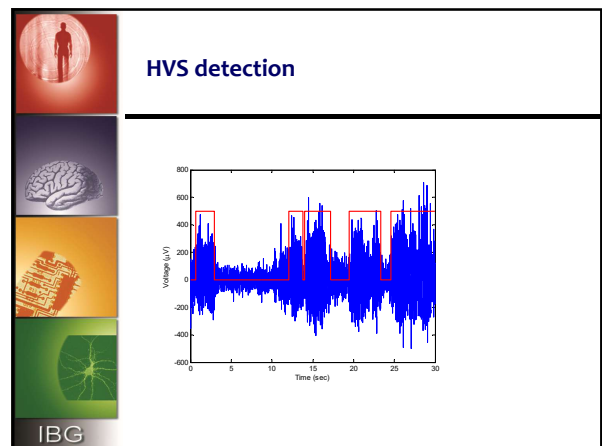
45



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